A2 123090612

Question 1

1

```
. //(1)
. //model 1
. reg wage yeduc nin
```

Source	SS	df	MS	Number of obs	=	61,136
				F(2, 61133)	=	4318.10
Model	3.2226e+14	2	1.6113e+14	Prob > F	=	0.0000
Residual	2.2812e+15	61,133	3.7315e+10	R-squared	=	0.1238
				Adj R-squared	=	0.1238
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	1.9e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	17826.42	193.9603	91.91	0.000	17446.26	18206.59
nin	-6880.068	714.7795	-9.63	0.000	-8281.038	-5479.099
_cons	72117.21	2519.881	28.62	0.000	67178.24	77056.19

- $\hat{\beta}_1 = 17826.42$, std.err is 193.9603
  - On average, an additional year of education **increases** monthly wage by 17826.42
  - p-value of $\hat{\beta}_1$ is: 0.000
- $\hat{\beta}_2 = -6880.068$, std.err is 714.7795
  - On average, an additional year of inpres school per children **decreases** monthly wage by 6880.068
  - p-value of $\hat{\beta}_2$ is: 0.000
- $\hat{\alpha} = 72117.21$, std.err is 2519.881
  - p-value of $\hat{\alpha}$ is: 0.000

  $\therefore$ The regression estimates are statistically **significant**.

  The two variables have an important impact on the model.
- $R^2 = 0.1238$, which means that 12.38% of the variability in monthly wage can be explained by years of education & province-level number of inpres school per children.

2

```
. //(2)
. //model 2
. reg nin yeduc
```

Source	SS	df	MS	Number of obs	=	152,989
				F(1, 152987)	=	592.60
Model	762.477939	1	762.477939	Prob > F	=	0.0000
Residual	196843.137	152,987	1.28666578	R-squared	=	0.0039
				Adj R-squared	=	0.0039
Total	197605.615	152,988	1.29164127	Root MSE	=	1.1343

nin	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	-.017078	.0007015	-24.34	0.000	-.018453	-.015703
_cons	2.24942	.0063333	355.17	0.000	2.237006	2.261833

If using the model $\text{nin} = \delta_0 + \delta_1 \text{education} + v$:

According to the data, $\hat{\delta_0} = 2.24942$, $\hat{\delta_1} = -0.017078$

$\because$ p-value of $\hat{\delta_0}$ is 0.000 & p-value of $\hat{\delta_1}$ is 0.000

$\therefore$ The regression estimates are statistically **significant**.

$\therefore$ `nin` & `education` are positively correlated.

$\therefore \text{wage}_{ij} = \alpha + \beta_1 \text{education}_i + \beta_2 \text{nin}_j + u_i$

$\implies \text{wage} = \alpha + \beta_1 \text{education} + \beta_2(\delta_0 + \delta_1 \text{education} + v) + u_i$

$= (\alpha + \beta_2\delta_0) + (\beta_1 + \beta_2\delta_1)\text{education} + (u + \beta_2 v)$

$\therefore \text{Bias}(\tilde{\beta_1}) = \beta_2\tilde{\delta_1} = (-0.017078) \times (-6880.068) = 117.497801$

$\because \tilde{\delta_0} > 0, \tilde{\delta_1} < 0, \tilde{\beta_2} < 0$

$\therefore$ For the $\tilde{\beta_1}$, there would be a **positive** bias. The increase shows the direction is upward.

```
. reg wage yeduc
```

Source	SS	df	MS	Number of obs	=	61,136
				F(1, 61134)	=	8530.77
Model	3.1880e+14	1	3.1880e+14	Prob > F	=	0.0000
Residual	2.2846e+15	61,134	3.7371e+10	R-squared	=	0.1225
				Adj R-squared	=	0.1224
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	1.9e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	17910.03	193.9109	92.36	0.000	17529.96	18290.09
_cons	57405.06	2004.897	28.63	0.000	53475.45	61334.66

Because wage is negatively correlated,

when `yeduc` increases by unit, besides the impact of increasing `yeduc` on `wage`,

`nin` also decreases with the increase of x, which also increases `wage`.

$\therefore$ The positive impact of `nin` on wage has been wrongly attributed to `yeduc`.

$\therefore \tilde{\beta_1}$ is positively biased.

3

$\because \hat{\beta}_2 < 0, \text{Bias}(\tilde{\beta}_1) > 0$

$\therefore \text{Cov}(\text{education}_i, \text{nin}_j) < 0$, the direction of $\text{Cov}(\text{education}_i, \text{nin}_j)$ should be negative

```
. //(3)
. // Check the correlation between nin & yeduc
. correlate nin yeduc
(obs=152,989)
```

	nin	yeduc
nin	1.0000	
yeduc	-0.0621	1.0000

Based on data, the direction of $\text{Cov}(\text{education}_i, \text{nin}_j)$ is also negative

$\therefore$ The two results are consistent with each other.

4

Model 3: $\text{wage}_{ij} = \alpha + \beta_1 \text{education}_i + \beta_2 \text{nin}_j + \beta_3 \text{en71} + u_i$

```
. //(4)
. //model 3
. reg wage yeduc nin en71
```

Source	SS	df	MS		
Model	3.2216e+14	3	1.0739e+14	Prob > F	0.0000
Residual	2.2774e+15	60,932	3.7376e+10	R-squared	0.1239
Total	2.5996e+15	60,935	4.2661e+10	Root MSE	1.9e+05

Number of obs	=	60,936				
F(3, 60932)	=	2873.13				
Prob > F	=	0.0000				
R-squared	=	0.1239				
Adj R-squared	=	0.1239				
Root MSE	=	1.9e+05				

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	17704.64	197.781	89.52	0.000	17316.98	18092.29
nin	-7513.557	735.1522	-10.22	0.000	-8954.457	-6072.656
en71	35557.16	9684.733	3.67	0.000	16575.06	54539.27
_cons	68233.96	2736.359	24.94	0.000	62870.69	73597.23

5

① The null hypothesis: $H_0 : \beta_3 = 0$

② The alternative $H_1 : \beta_3 \neq 0$

③ Choose 5% as the significance value

④ $t_{\hat{\beta}_3} = \frac{\hat{\beta}_3}{se(\beta_3)} = \frac{35557.16}{9684.733} \approx 3.67$

⑤ $3.67 > 1.96$, the null hypothesis can be rejected because the t-statistic exceeds the critical value

∴ We can take the alternative hypothesis $H_1 : \beta_3 \neq 0$

∴ The enrollment rate in 1971 of each province does change the wage outcome.

6

Use **F-test**.

```
. //(6)
. //Find the joint significance of nin and en71
. test nin en71

 ( 1)  nin = 0
 ( 2)  en71 = 0

       F(  2, 60932) =   53.17
            Prob > F =    0.0000
```

The null hypothesis: $H_0 : \beta_2 = \beta_3 = 0$, coefficients pf nin & en71 are all 0

the alternative $H_1 : \beta_2 \neq 0$ or $\beta_3 \neq 0$

Choose 1% as the significance value

Through model $\text{wage}_{ij} = \beta_0 + \beta_1 \text{yeduc}_i + \beta_2 \text{nin}_j + \beta_3 \text{en71} + u_i \implies \text{SSR}_{ur}$

Through wage $= \beta_0 + \beta_1 \text{yeduc} + u \implies \text{SSR}_r$

$F = \frac{[SSR_r - SSR_{ur}]/q}{SSR_{ur}/n-k-1} = 96.3246$

$F = \frac{[SSR_r - SSR_{ur}]/q}{SSR_{ur}/n-k-1} \sim F_{2,60932}$, the critical value $c = 53.17$

$\because F > 53.17$

∴ We can reject H_0

∴ `nin` & `en71` jointly affect wage, the effects of `nin` & `en71` are significant.

Question 2

1

```
. //linear model
. reg high_wage yeduc nin
```

Source	SS	df	MS	Number of obs	=	61,136
				F(2, 61133)	=	5721.30
Model	2409.73024	2	1204.86512	Prob > F	=	0.0000
Residual	12874.1651	61,133	.210592725	R-squared	=	0.1577
				Adj R-squared	=	0.1576
Total	15283.8953	61,135	.250002377	Root MSE	=	.4589

high_wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	.0489753	.0004608	106.29	0.000	.0480722	.0498785
nin	-.0123811	.0016981	-7.29	0.000	-.0157093	-.0090529
_cons	.0600717	.0059863	10.03	0.000	.0483385	.071805

```
. //probit model
. probit high_wage yeduc nin

Iteration 0:  Log likelihood = -42376.037
Iteration 1:  Log likelihood = -37223.171
Iteration 2:  Log likelihood = -37222.747
Iteration 3:  Log likelihood = -37222.747
```

Probit regression — Number of obs = 61,136; LR chi2(2) = 10306.58; Prob > chi2 = 0.0000; Pseudo R2 = 0.1216

Log likelihood = -37222.747

high_wage	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
yeduc	.1370616	.0014279	95.99	0.000	.1342629	.1398603
nin	-.0346085	.004821	-7.18	0.000	-.0440573	-.0251596
_cons	-1.234311	.017857	-69.12	0.000	-1.26931	-1.199312

Table:

Dependent variable: high_wage = 1 if wage ≥ 19200, = 0 if wage < 19200

regression model	LPM	Probit
yeduc	0.0489753 (0.0004608)	0.1370616 (0.0014279)
nin	-0.0123811 (0.0016981)	-0.0346085 (0.004821)
constant	0.0600717 (0.0059863)	-1.234311 (0.017857)

- The coefficient of `yeduc` in LPM means that:

- for each additional year of education
- holding other factors constant
- the probability of having a high wage ≥ 19200 would increase by about 4.9%

2

Using a linear probability model (LPM) for a binary outcome variable Y instead of a non-linear model like the Probit model has several drawbacks:

1. **Predicted Probability Bounds**: In a linear probability model, there is no restriction to keep the predicted probabilities within the range of 0 and 1. This can lead to unrealistic predictions (i.e., probabilities less than 0 or greater than 1), which are not meaningful in a probability context. Non-linear models, such as the Probit model, inherently restrict predictions within this range.
2. **Heteroscedasticity**: The variance of the error term in a linear probability model is not constant; it depends on the values of the predictors. This issue, known as heteroscedasticity, can lead to inefficient estimates and invalid standard errors, making hypothesis testing less reliable. Non-linear models, like the Probit, do not suffer from this issue as they assume a constant variance in the latent error term.
3. **Non-linearity of Relationships**: LPM assumes a linear relationship between the predictors and the probability of the outcome occurring. This assumption may not hold in many cases, as the effect of the predictors on the probability of an event is often non-linear, especially as probabilities approach 0 or 1. Non-linear models, like Probit, are more flexible in capturing such non-linear relationships.

3

```
. //probit model
. probit high_wage yeduc nin

Iteration 0:  Log likelihood = -42376.037
Iteration 1:  Log likelihood = -37223.171
Iteration 2:  Log likelihood = -37222.747
Iteration 3:  Log likelihood = -37222.747

Probit regression                                 Number of obs =     61,136
                                                  LR chi2(2)    =   10306.58
                                                  Prob > chi2   =     0.0000
Log likelihood = -37222.747                       Pseudo R2     =     0.1216
```

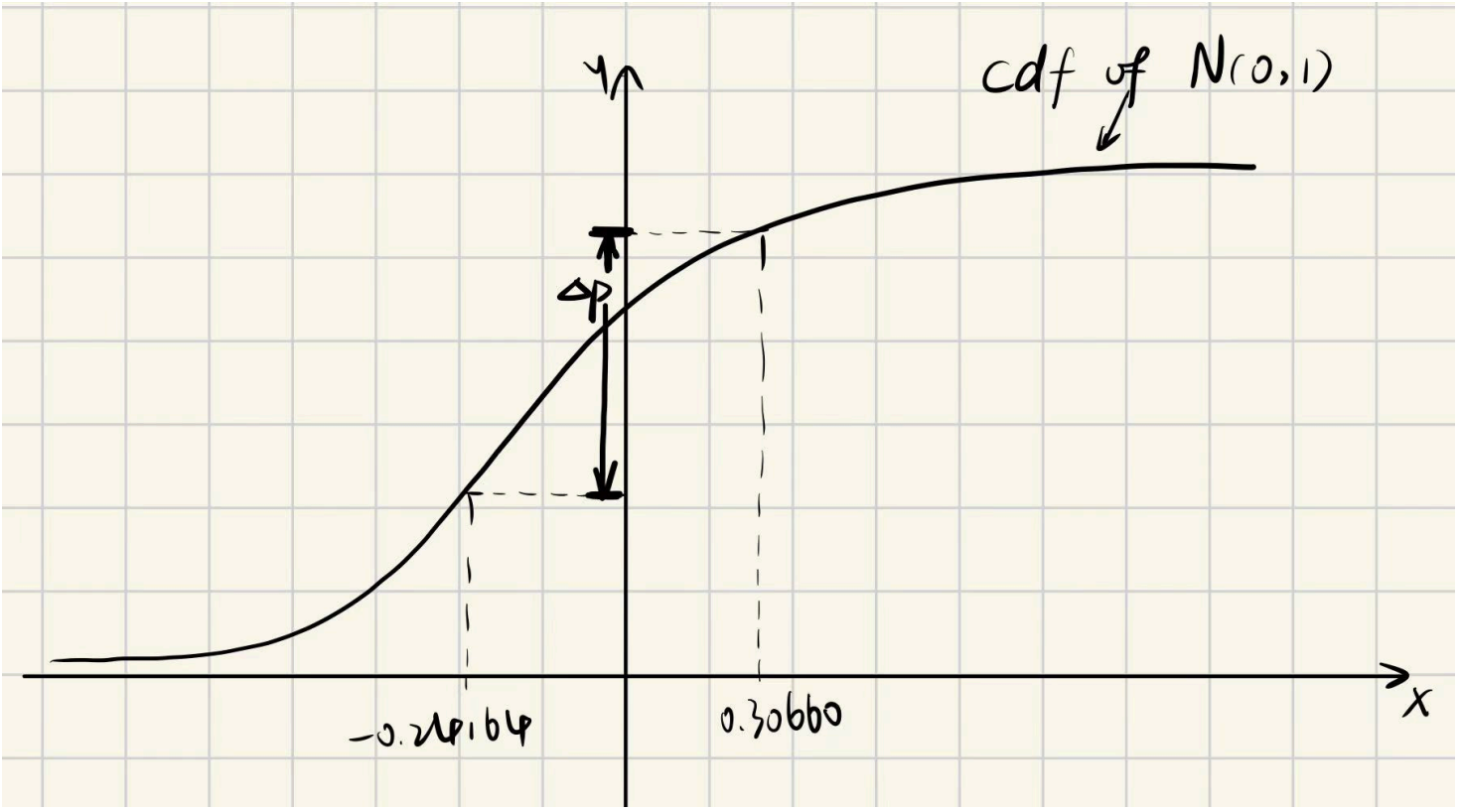
high_wage	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
yeduc	.1370616	.0014279	95.99	0.000	.1342629	.1398603
nin	-.0346085	.004821	-7.18	0.000	-.0440573	-.0251596
_cons	-1.234311	.017857	-69.12	0.000	-1.26931	-1.199312

$\hat{\beta}_1 = 0.1370616, \hat{\beta}_2 = -0.0346085, \hat{\beta}_0 = -1.234311$
$P(Y_i = 1|X = 8, Z = 3) = \Phi(\hat{\beta}_0 + 8\hat{\beta}_1 + 3\hat{\beta}_2) = \Phi(-0.24164)$

$P(Y_i = 1|X = 12, Z = 3) = \Phi(\hat{\beta}_0 + 12\hat{\beta}_1 + 3\hat{\beta}_2) = \Phi(0.30660)$

$\Delta P = \Phi(0.30660) - \Phi(-0.24164)$

4



Question 3

1

```
. //Question 3
. //(1)
. //Generate high_educ
. gen high_level = .
(152,990 missing values generated)

. replace high_level = 1 if yeduc >= 9 & !missing(wage)
(38,650 real changes made)

. replace high_level = 0 if yeduc < 9
(82,185 real changes made)

.
. //Show the relationship between wage & high level
. reg wage high_level
```

Source	SS	df	MS	Number of obs	=	61,136
				F(1, 61134)	=	4561.41
Model	1.8076e+14	1	1.8076e+14	Prob > F	=	0.0000
Residual	2.4227e+15	61,134	3.9629e+10	R-squared	=	0.0694
				Adj R-squared	=	0.0694
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	2.0e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
high_level	112764.3	1669.638	67.54	0.000	109491.8	116036.8
_cons	156631.9	1327.543	117.99	0.000	154029.9	159233.9

- The coefficient of high level is 112764.3
  - which means, individuals in high-education group (`high_level` = 1) earn on average $112764.3 more than low-education group (`high_level` = 0), holding all else constant.

2

(1)

wage $= \beta_0 + \beta_1$ high*level* + $\beta{2}\text{urban}+\beta_{3}\text{urban} \cdot$ high_level $+u$

```
.
. //(2)
. //Generate urbanhigh_level
. gen interaction = urban * high_level
(32,155 missing values generated)

.
. //Regress in the interaction term
. reg wage high_level urban interaction
```

Source	SS	df	MS	Number of obs	=	61,136
				F(3, 61132)	=	1727.11
Model	2.0342e+14	3	6.7805e+13	Prob > F	=	0.0000
Residual	2.4000e+15	61,132	3.9259e+10	R-squared	=	0.0781
				Adj R-squared	=	0.0781
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	2.0e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
high_level	99802.76	1914.016	52.14	0.000	96051.29	103554.2
urban	25014.22	3553.394	7.04	0.000	18049.56	31978.88
interaction	23411.8	4131.893	5.67	0.000	15313.28	31510.32
_cons	152485.9	1446.661	105.41	0.000	149650.4	155321.3

- $\hat{\beta_0} = 152485.9$
- $\hat{\beta_1} = 99802.76$
  - This is the extra monthly wage on average for those having high education & born in rural place
- $\hat{\beta_2} = 25014.22$
  - This is the difference of monthly wage on average between urban born people with low education level & rural born people with low education level
- $\hat{\beta_3} = 23411.8$
  - This is the extra monthly wage on average for urban born people with high education level compared with rural born people with low education level

(2)

- Null hypothesis: $H_0 : \beta_3 = 0$
  - Birthplace can't affect the slope of high-level on wage
- The alternative hypothesis: $H_1 : \beta_3 \neq 0$
  - Birthplace can affect the slope of high-level on wage

  Choose 5% as the significance value.

  $\because$ p-value for the interaction term is $0.0000 < 0.05$

  $\therefore$ We can reject H_0

  $\therefore$ Interaction between high education level & urban birthplace is statistically significant.

  $\therefore$ Birthplace can affect the slope of high-level on wage.

Do file

```
clear

use "D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A2\inpresdata.dta", clear

//Question 1

//(1)
//model 1
reg wage yeduc nin

//(2)
//model 2
reg nin yeduc
reg wage yeduc

//(3)
// Check the correlation between nin & yeduc
correlate nin yeduc

//(4)
//model 3
reg wage yeduc nin en71

//(6)
//Find the joint significance of nin and en71
test nin en71

//---
//Question 2

//(1)
gen high_wage = .
replace high_wage = 1 if wage >= 192000 & !missing(wage)
replace high_wage = 0 if wage < 192000

//linear model
reg high_wage yeduc nin

//probit model
probit high_wage yeduc nin

//---
//Question 3
//(1)
//Generate high_educ
gen high_level = .
replace high_level = 1 if yeduc >= 9 & !missing(wage)
replace high_level = 0 if yeduc < 9

//Show the relationship between wage & high level
reg wage high_level

//(2)
```

```
54 //Generate urbanhigh_level
55 gen interaction = urban * high_level
56
57 //Regress in the interaction term
58 reg wage high_level urban interaction
59
60 //Check if the urban will affect the slope of high_level
61 reg urban interaction
```

Log file

```
------------------------------------------------------------------------------------------------------------------
      name:  <unnamed>
       log:  D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A2\A2.log
  log type:  text
 opened on:   1 Nov 2024, 16:15:44

. do "D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A2\A2.do"

. clear

.
. use "D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A2\inpresdata.dta", clear

.
. //Question 1
.
. //(1)
. //model 1
. reg wage yeduc nin

      Source |       SS           df       MS      Number of obs   =    61,136
-------------+----------------------------------   F(2, 61133)     =   4318.10
       Model |  3.2226e+14         2  1.6113e+14   Prob > F        =    0.0000
    Residual |  2.2812e+15    61,133  3.7315e+10   R-squared       =    0.1238
-------------+----------------------------------   Adj R-squared   =    0.1238
       Total |  2.6034e+15    61,135  4.2585e+10   Root MSE        =    1.9e+05

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       yeduc |   17826.42   193.9603    91.91   0.000     17446.26    18206.59
         nin |  -6880.068   714.7795    -9.63   0.000    -8281.038   -5479.099
       _cons |   72117.21   2519.881    28.62   0.000     67178.24    77056.19
------------------------------------------------------------------------------

.
. //(2)
. //model 2
. reg nin yeduc

      Source |       SS           df       MS      Number of obs   =   152,989
-------------+----------------------------------   F(1, 152987)    =    592.60
       Model |  762.477939         1  762.477939   Prob > F        =    0.0000
    Residual |  196843.137   152,987  1.28666578   R-squared       =    0.0039
-------------+----------------------------------   Adj R-squared   =    0.0039
       Total |  197605.615   152,988  1.29164127   Root MSE        =    1.1343

------------------------------------------------------------------------------
         nin | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       yeduc |   -.017078   .0007015   -24.34   0.000     -.018453    -.015703
       _cons |    2.24942   .0063333   355.17   0.000     2.237006    2.261833
------------------------------------------------------------------------------

. reg wage yeduc

      Source |       SS           df       MS      Number of obs   =    61,136
-------------+----------------------------------   F(1, 61134)     =   8530.77
       Model |  3.1880e+14         1  3.1880e+14   Prob > F        =    0.0000
    Residual |  2.2846e+15    61,134  3.7371e+10   R-squared       =    0.1225
-------------+----------------------------------   Adj R-squared   =    0.1224
       Total |  2.6034e+15    61,135  4.2585e+10   Root MSE        =    1.9e+05

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       yeduc |   17910.03   193.9109    92.36   0.000     17529.96    18290.09
       _cons |   57405.06   2004.897    28.63   0.000     53475.45    61334.66
------------------------------------------------------------------------------

.
. //(3)
. // Check the correlation between nin & yeduc
```

```
. correlate nin yeduc
(obs=152,989)

             |      nin    yeduc
-------------+------------------
         nin |   1.0000
       yeduc |  -0.0621   1.0000


.
. //(4)
. //model 3
. reg wage yeduc nin en71

      Source |       SS           df       MS      Number of obs   =    60,936
-------------+----------------------------------   F(3, 60932)     =   2873.13
       Model |  3.2216e+14         3  1.0739e+14   Prob > F        =    0.0000
    Residual |  2.2774e+15    60,932  3.7376e+10   R-squared       =    0.1239
-------------+----------------------------------   Adj R-squared   =    0.1239
       Total |  2.5996e+15    60,935  4.2661e+10   Root MSE        =    1.9e+05

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       yeduc |   17704.64    197.781    89.52   0.000     17316.98    18092.29
         nin |  -7513.557   735.1522   -10.22   0.000    -8954.457   -6072.656
        en71 |   35557.16   9684.733     3.67   0.000     16575.06    54539.27
       _cons |   68233.96   2736.359    24.94   0.000     62870.69    73597.23
------------------------------------------------------------------------------

.
. //(6)
. //Find the joint significance of nin and en71
. test nin en71

 ( 1)  nin = 0
 ( 2)  en71 = 0

       F(  2, 60932) =   53.17
            Prob > F =    0.0000

.
. //---
. //Question 2
.
. //(1)
. gen high_wage = .
(152,990 missing values generated)

. replace high_wage = 1 if wage >= 192000 & !missing(wage)
(30,648 real changes made)

. replace high_wage = 0 if wage < 192000
(30,488 real changes made)

.
. //linear model
. reg high_wage yeduc nin

      Source |       SS           df       MS      Number of obs   =    61,136
-------------+----------------------------------   F(2, 61133)     =   5721.30
       Model |  2409.73024         2  1204.86512   Prob > F        =    0.0000
    Residual |  12874.1651    61,133  .210592725   R-squared       =    0.1577
-------------+----------------------------------   Adj R-squared   =    0.1576
       Total |  15283.8953    61,135  .250002377   Root MSE        =     .4589

------------------------------------------------------------------------------
   high_wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       yeduc |   .0489753   .0004608   106.29   0.000     .0480722    .0498785
         nin |  -.0123811   .0016981    -7.29   0.000    -.0157093   -.0090529
       _cons |   .0600717   .0059863    10.03   0.000     .0483385     .071805
------------------------------------------------------------------------------

.
. //probit model
. probit high_wage yeduc nin
```

```
Iteration 0:  Log likelihood = -42376.037
Iteration 1:  Log likelihood = -37223.171
Iteration 2:  Log likelihood = -37222.747
Iteration 3:  Log likelihood = -37222.747

Probit regression                                 Number of obs =   61,136
                                                  LR chi2(2)    = 10306.58
                                                  Prob > chi2   =   0.0000
Log likelihood = -37222.747                       Pseudo R2     =   0.1216

------------------------------------------------------------------------------
   high_wage | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       yeduc |   .1370616   .0014279    95.99   0.000     .1342629    .1398603
         nin |  -.0346085    .004821    -7.18   0.000    -.0440573   -.0251596
       _cons |  -1.234311    .017857   -69.12   0.000     -1.26931   -1.199312
------------------------------------------------------------------------------

.
. //---
. //Question 3
. //(1)
. //Generate high_educ
. gen high_level = .
(152,990 missing values generated)

. replace high_level = 1 if yeduc >= 9 & !missing(wage)
(38,650 real changes made)

. replace high_level = 0 if yeduc < 9
(82,185 real changes made)

.
. //Show the relationship between wage & high level
. reg wage high_level

      Source |       SS           df       MS      Number of obs   =    61,136
-------------+----------------------------------   F(1, 61134)     =   4561.41
       Model |  1.8076e+14         1  1.8076e+14   Prob > F        =    0.0000
    Residual |  2.4227e+15    61,134  3.9629e+10   R-squared       =    0.0694
-------------+----------------------------------   Adj R-squared   =    0.0694
       Total |  2.6034e+15    61,135  4.2585e+10   Root MSE        =    2.0e+05

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  high_level |   112764.3   1669.638    67.54   0.000     109491.8    116036.8
       _cons |   156631.9   1327.543   117.99   0.000     154029.9    159233.9
------------------------------------------------------------------------------

.
. //(2)
. //Generate urbanhigh_level
. gen interaction = urban * high_level
(32,155 missing values generated)

.
. //Regress in the interaction term
. reg wage high_level urban interaction

      Source |       SS           df       MS      Number of obs   =    61,136
-------------+----------------------------------   F(3, 61132)     =   1727.11
       Model |  2.0342e+14         3  6.7805e+13   Prob > F        =    0.0000
    Residual |  2.4000e+15    61,132  3.9259e+10   R-squared       =    0.0781
-------------+----------------------------------   Adj R-squared   =    0.0781
       Total |  2.6034e+15    61,135  4.2585e+10   Root MSE        =    2.0e+05

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  high_level |   99802.76   1914.016    52.14   0.000     96051.29    103554.2
       urban |   25014.22   3553.394     7.04   0.000     18049.56    31978.88
 interaction |    23411.8   4131.893     5.67   0.000     15313.28    31510.32
       _cons |   152485.9   1446.661   105.41   0.000     149650.4    155321.3
------------------------------------------------------------------------------
```

```
. //Check if the urban will affect the slope of high_level
. reg urban interaction

      Source |       SS           df       MS      Number of obs   =   120,835
-------------+----------------------------------   F(1, 120833)    >  99999.00
       Model |    10264.07         1    10264.07   Prob > F        =    0.0000
    Residual |   7835.1608   120,833  .064842889   R-squared       =    0.5671
-------------+----------------------------------   Adj R-squared   =    0.5671
       Total |  18099.2308   120,834  .149785911   Root MSE        =    .25464

------------------------------------------------------------------------------
       urban | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 interaction |   .9205923   .0023139   397.86   0.000     .9160571    .9251274
       _cons |   .0794077   .0007778   102.09   0.000     .0778832    .0809322
------------------------------------------------------------------------------

.
end of do-file

. log close
      name:  <unnamed>
       log:  D:\MG\! CUHKSZ\~! 大二上\ECO3121\Assignment\A2\A2.log
  log type:  text
 closed on:   1 Nov 2024, 16:16:01
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```